SE LAB7

NAME: HIMANI PRABHU SECTION : D SRN: PES2UG24CS194

1. Objective

The objective of this lab is to apply fuzz testing to three Python functions that process season-pass dates, trail gradients, and lift-access zone codes. The starter application works with ordinary valid examples, but it does not safely handle every possible input. Hypothesis is used with pytest to generate a wide range of inputs and reveal cases that cause the functions to raise exceptions.

2. Tools and Environment

Python 3.13.15

pytest 9.2.0

Hypothesis 6.98.9

Visual Studio Code and Windows PowerShell

A project-local virtual environment named .venv

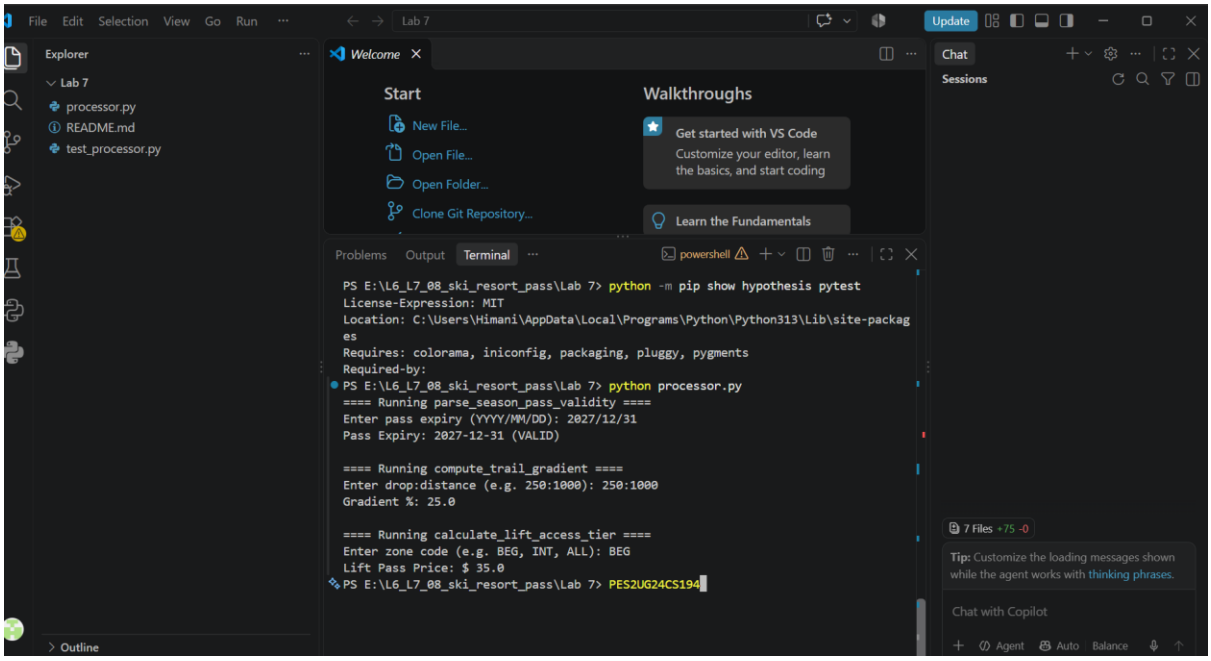
A virtual environment was used to isolate the lab dependencies. The initially installed Hypothesis build encountered a Windows Application Control restriction when loading its native component. The lab was then run successfully with Hypothesis 6.98.9 inside .venv.

3. Application Functions

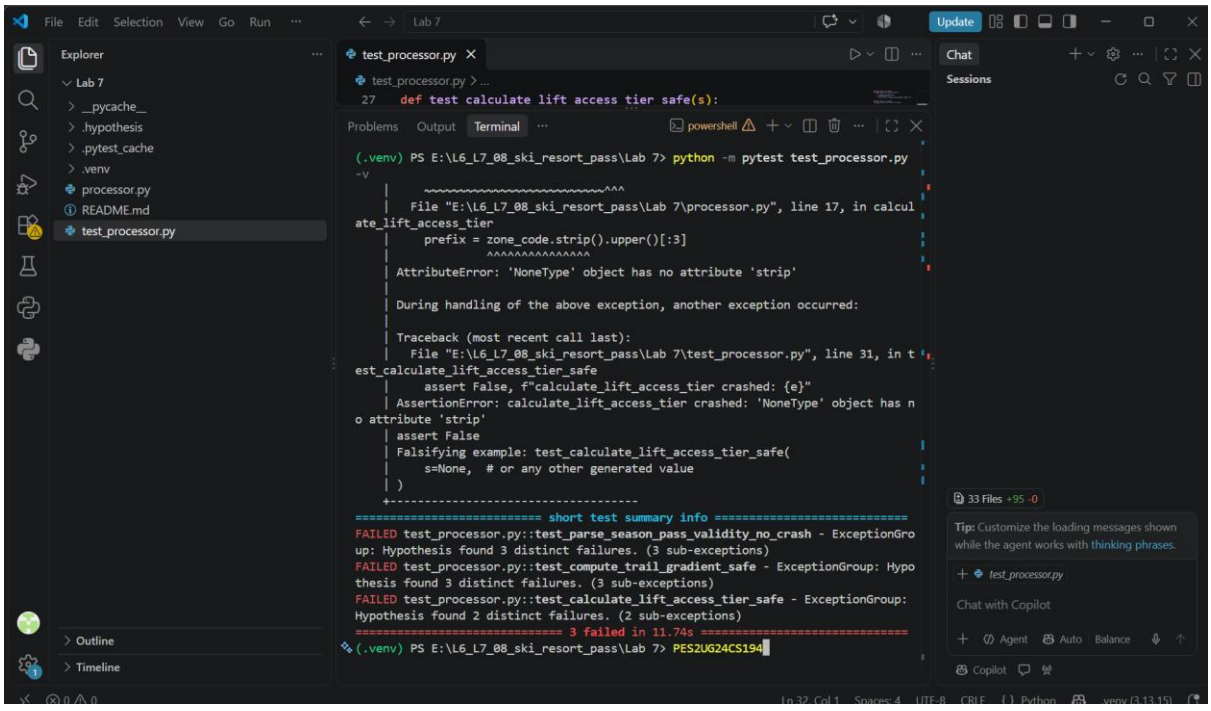
Function	Purpose	Expected valid example
parse_season_pass_validity(date_str)	Parses a YYYY/MM/DD expiry date and determines whether the year is 2026 or later.	2027/12/31 → VALID
compute_trail_gradient(elevation_str)	Reads drop:distance and calculates the gradient percentage.	250:1000 → 25.0
calculate_lift_access_tier(zone_code)	Looks up the price associated with a lift-access zone prefix.	BEG → $35.0

SCREENSHOT EVIDENCE

01_Original_Application.png



02_Fuzz_Test_Failures.png



03_Fixed_Application.png

```
        assert False, f"calculate_lift_access_tier crashed: {e}"
(.venv) PS E:\L6_L7_08_ski_resort_pass\Lab 7> python processor.py
==== Running parse_season_pass_validity ====
Enter pass expiry (YYYY/MM/DD): 2027/12/31
Pass Expiry: 2027-12-31 (VALID)

==== Running compute_trail_gradient ====
Enter drop:distance (e.g. 250:1000): 250:1000
Gradient %: 25.0

==== Running calculate_lift_access_tier ====
Enter zone code (e.g. BEG, INT, ALL): BEG
Lift Pass Price: $ 35.0
(.venv) PS E:\L6_L7_08_ski_resort_pass\Lab 7> PES2UG24CS194
```

04_Passing_Fuzz_Tests.png

```
==== Running calculate_lift_access_tier ====
Enter zone code (e.g. BEG, INT, ALL): BEG
Lift Pass Price: $ 35.0
(.venv) PS E:\L6_L7_08_ski_resort_pass\Lab 7> python -m pytest test_processor.py
-v
============================= test session starts ==============================
platform win32 -- Python 3.13.15, pytest-9.2.0, pluggy-1.7.0 -- E:\L6_L7_08_ski_r
esort_pass\Lab 7\.venv\Scripts\python.exe
cachedir: .pytest_cache
hypothesis profile 'default' -> database=DirectoryBasedExampleDatabase(WindowsPat
h('E:/L6_L7_08_ski_resort_pass/Lab 7/.hypothesis/examples'))
rootdir: E:\L6_L7_08_ski_resort_pass\Lab 7
plugins: hypothesis-6.98.9
collected 3 items

test_processor.py::test_parse_season_pass_validity_no_crash PASSED        [ 33%]
test_processor.py::test_compute_trail_gradient_safe PASSED                [ 66%]
test_processor.py::test_calculate_lift_access_tier_safe PASSED            [100%]

============================== 3 passed in 1.72s ===============================
(.venv) PS E:\L6_L7_08_ski_resort_pass\Lab 7> PES2UG24CS194
```

8. Conclusion

This lab demonstrated how fuzz testing complements ordinary example-based testing. The original Ski Resort Pass functions handled the selected valid examples but failed for some unexpected inputs generated by Hypothesis. Defensive checks, validation of parsed values, safe dictionary lookup, and controlled fallback results were added. The final Hypothesis run passed all three tests, showing that the tested functions no longer raise exceptions for the generated text and None inputs.

The tests in this lab specifically verify crash resistance for the generated input strategy. Additional assertions and strategies can be added in future tests to verify exact return values, date boundaries, numeric ranges, and all supported zone codes.